A MATHEMATICAL THEORY OF CONVECTION

By Victor P. Starr

Massachusetts Institute of Technology[1]

(Manuscript received 1 October 1948)

ABSTRACT

Since the motions of the atmosphere involve the presence of external heat sources and heat sinks, it is important to incorporate these effects into the hydrodynamical theory of air motions. In the discussion contained in this paper a method for accomplishing such a synthesis of hydrodynamics and thermodynamics is formulated for the case of two-dimensional motion. The use of the method is illustrated by an exact integration of the equations for the case of a circular vortex. Possible further elaborations of the method are discussed with a view of throwing light upon the fundamental nature of the general circulation of the atmosphere.

1. Introduction

It is recognized by meteorologists that, in order to obtain a complete dynamic picture of at least the large-scale motions in the atmosphere, it is necessary to give proper consideration to the fact that, generally speaking, the air motions are accompanied by external heat exchanges through radiation and otherwise. We shall here use the word *convection* to denote such fluid motions. It might appear that the analytical study of the motions of a thermally active fluid is a subject of very great complexity. However, this is not always the case, and in what follows a certain simple approach to the more elementary problems of this type is given. A review of the literature on this subject has been given recently by Hicks [2].

2. Mode of attack

Let us consider the steady two-dimensional motions in the x, y-plane of a fluid which obeys the simple equations of motion

$$\rho \frac{du}{dt} = -\frac{\partial p}{\partial x}; \qquad \rho \frac{dv}{dt} = -\frac{\partial p}{\partial y}, \tag{1}$$

and the continuity equation

$$\partial \rho u/\partial x + \partial \rho v/\partial y = 0. \tag{2}$$

In (1) and (2) u, v are the particle velocity components in the x- and y-directions respectively, ρ is the density, p is pressure and t is time. By virtue of (2) there exists a stream function ψ such that

$$u = -\alpha\, \partial\psi/\partial y; \qquad v = \alpha\, \partial\psi/\partial x, \tag{3}$$

where α is the specific volume, that is, the reciprocal of the density ρ. In what follows we consider the quantities u, v, p, α, ρ all to be functions of the (cartesian) coordinates x and y.

We now proceed as follows. The pressure p may be eliminated from equations (1) by cross differentiation and subtraction, and then u and v may be eliminated from the result by means of (3), so that finally a single equation involving only α and ψ may be obtained. This equation is

$$\begin{aligned}\psi_x\psi_y\alpha_{xx} - (\psi_x^2 - \psi_y^2)\alpha_{xy} - \psi_x\psi_y\alpha_{yy} \\ = 2\psi_x(\psi_{xx} + \psi_{yy})\alpha_y - 2\psi_y(\psi_{xx} + \psi_{yy})\alpha_x \\ + [\psi_x(\psi_{yyy} + \psi_{xxy}) - \psi_y(\psi_{xxx} + \psi_{xyy})]\alpha.\end{aligned} \tag{4}$$

The subscripts are used to denote partial differentiations with respect to the designated variables. If we regard α as a given function, then (4) is the differential equation to be satisfied by ψ. As an equation for ψ, (4) is seen to be of the third order and nonlinear, so that solutions can be readily obtained only for very simple choices of α. We shall therefore not consider this procedure further.

On the other hand, if we regard ψ as being a given function, then (4) becomes a partial differential equation for α. So regarded, it is seen that (4) *is an equation of second order which is linear*. Further examination reveals that it is of the *hyperbolic type*, and that *the characteristics are the streamlines and their orthogonal trajectories*, whatever be the given function ψ (see for example, Webster [6]). According to the theory of differential equations, the solutions of (4) may involve arbitrary functions. It is therefore seen that for one and the same field of ψ, various fields of α may exist. Indeed, generally speaking, we could consider the Cauchy problem corresponding to (4) in which we have given, in addition to the field of ψ, the value of α and its normal derivative along a given torsion strip not coinciding with the characteristics.

Once a consistent pair of functions ψ and α is found, the particle velocities u and v become known from (3),

[1] This research was performed under contract with the Watson Laboratories of the U. S. Air Forces.

Unauthenticated | Downloaded 01/27/26 08:09 PM UTC

and the pressure p may be obtained from (1), since (4) is the condition of integrability for the relation

$$dp = \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy, \tag{5}$$

when the values of $\partial p/\partial x$ and $\partial p/\partial y$ are given by (1).

It is to be noted that the manipulations indicated above are independent of the thermodynamic properties of the fluid considered. Thus we may, if we so desire, at this point assume that the fluid is compressible in the sense that it obeys the ideal equation of state, namely,

$$T = A\alpha p, \tag{6}$$

where $1/A$ is the gas constant and T is the absolute temperature which thus becomes a known function of x and y. We may also assume an appropriate form for the first law of thermodynamics such as the one for an ideal gas, namely,

$$\frac{dq}{dt} = c_v \frac{dT}{dt} + p \frac{d\alpha}{dt}, \tag{7}$$

where q is heat energy in mechanical units and c_v is the specific heat at constant volume. In view of the fact that all the variables on the right-hand side of (7) are known, we may compute the distribution of the heating and cooling per unit volume, namely, $\rho\, dq/dt$.

We thus arrive at the required heating and cooling necessary to maintain the fixed temperature distribution against advection *as a result* of the computation, the procedure being in this sense an indirect one.

3. A simple example

In order to illustrate the method by means of a concrete example, let us assume that ψ is given by[2]

$$\psi = \tfrac{1}{2}\kappa(x^2 + y^2), \tag{8}$$

where κ is a positive constant so that the motion is counterclockwise in concentric circles, although the particle speed $c \equiv \sqrt{u^2 + v^2}$ may vary along the streamlines. In order to exclude all infinities and singularities later, it will be assumed also that the region to be considered is bounded by circular walls at radii $r = r_1$ internally and at $r = r_2$ externally. The fluid may thus be thought of as contained in an annular box with plane top and bottom and cylindrical vertical sides.

It is simple to verify that equation (4) now becomes

$$xy\alpha_{xx} - (x^2 - y^2)\alpha_{xy} - xy\alpha_{yy} = 4(x\alpha_y - y\alpha_x). \tag{9}$$

The differential equation (9) yields readily to Monge's method of integration (see, for example, Forsyth [1]) and the process gives two intermediary integrals, namely,

$$\begin{aligned} x\alpha_x + y\alpha_y &= F_1(x^2 + y^2) - 3\alpha \\ y\alpha_x - x\alpha_y &= x^{-3}F_2(y/x), \end{aligned} \tag{10}$$

where F_1 and F_2 indicate arbitrary functions. From these it follows that the general integral of (9) is

$$\alpha = \phi_1(x^2 + y^2) + (x^2 + y^2)^{-\frac{3}{2}}\phi_2(y/x), \tag{11}$$

in which ϕ_1 and ϕ_2 are arbitrary functions related to F_1 and F_2.

For simplicity the solution (11) will be particularized by taking

$$\alpha = \alpha_0 + \epsilon r^{-3} \cos\theta, \tag{12}$$

in which α_0 and ϵ are positive constants and

$$r = \sqrt{x^2 + y^2}; \qquad \theta = \tan^{-1}(y/x). \tag{13}$$

The velocity components u and v may now be written explicitly, and in terms of the polar coordinates r, θ it is found that

$$c \equiv \sqrt{u^2 + v^2} = \kappa(\alpha_0 + \epsilon r^{-3} \cos\theta)r. \tag{14}$$

The pressure distribution is best found from the polar form of the equations (1) which, in the present case, are

$$\frac{\partial p}{\partial r} = \rho \frac{c^2}{r}; \qquad \frac{\partial p}{\partial \theta} = -\rho c \frac{\partial c}{\partial \theta}. \tag{15}$$

Substitution from (12), (14) and integration gives that

$$p = \kappa^2(\tfrac{1}{2}\alpha_0 - \epsilon r^{-3} \cos\theta)r^2 + p_0, \tag{16}$$

in which p_0 is an arbitrary constant.

Using the form of the equation of state (6), the temperature distribution is

$$T = A\kappa^2(\tfrac{1}{2}\alpha_0^2 - \tfrac{1}{2}\alpha_0\epsilon r^{-3} \cos\theta - \epsilon^2 r^{-6} \cos^2\theta)r^2 + Ap_0(\alpha_0 + \epsilon r^{-3} \cos\theta). \tag{17}$$

Since it follows that

$$\frac{d\alpha}{dt} = \frac{c}{r}\frac{\partial \alpha}{\partial \theta} = -\kappa\left(\alpha_0 + \frac{\epsilon}{r^3}\cos\theta\right)\frac{\epsilon}{r^3}\sin\theta, \tag{18}$$

and that

$$\frac{dT}{dt} = \frac{c}{r}\frac{\partial T}{\partial \theta} = A\kappa\left(\alpha_0 + \frac{\epsilon}{r^3}\cos\theta\right)\left[\kappa^2\left(\frac{\alpha_0\epsilon}{2r}\sin\theta + 2\frac{\epsilon^2}{r^4}\sin\theta\cos\theta\right) - p_0\frac{\epsilon}{r^3}\sin\theta\right], \tag{19}$$

we may state with the aid of (7) that the heating per

[2] This choice of the function ψ is one of a general class which is characterized by a constancy of the *momentum* along each individual streamline, *i.e.*,

$$\psi_x^2 + \psi_y^2 = \phi(\psi),$$

where $\phi(\psi)$ is an arbitrary function. It has been shown by G. W. Platzman (unpublished) that this condition together with (4) forms a system which may be integrated in its generality so that a result embracing all such cases is obtained. This solution is obtained through the use of intrinsic coordinates ξ and η taken along the streamlines and their orthogonal trajectories respectively.

Unauthenticated | Downloaded 01/27/26 08:09 PM UTC

unit volume is given by

$$\rho \frac{dq}{dt} = c_v A \kappa \left[\kappa^2 \left(\frac{\alpha_0 \epsilon}{2r} + \frac{2\epsilon^2}{r^4} \cos\theta \right) - \frac{p_0 \epsilon}{r^3} \right] \sin\theta - \frac{\kappa\epsilon}{r^3} \left[\kappa^2 \left(\frac{\alpha_0}{2} - \frac{\epsilon}{r^3} \cos\theta \right) r^2 + p_0 \right] \sin\theta. \tag{20}$$

A computed diagram of the illustrative model is given in fig. 1. The values of the constants used are:

A =	0.3484×10^{-6}	deg sec^2 cm^{-2}
κ =	5×10^{-8}	gm cm^{-3} sec^{-1}
ϵ =	10^{24}	cm^6 gm^{-1}
r_1 =	2×10^7	cm
r_2 =	5×10^7	cm
p_0 =	10^6	gm cm^{-1} sec^{-2}
α_0 =	10^3	cm^3 gm^{-1}
c_v =	0.7146×10^7	cm^2 deg^{-1} sec^{-2}

4. Physical properties of illustrative model

The model described above possesses a number of physical characteristics which warrant discussion. In the first place it is to be noted that the expression for the heating given by (20) is antisymmetrical about the horizontal axis of fig. 1, so that as much heat energy is lost in the upper half of the system as is gained in the lower half, the system thus being in accord with the principle of conservation of energy. Furthermore, a reversal of the motions (a change in the sign of κ) would merely reverse the sign of the heating and not alter the other characteristic physical features, so that the system is a reversible one as should follow independently from the fact that no dissipative processes such as friction have been assumed.

The particle speed c along any given streamline, as given by (14), is greatest at $\theta = 0$, that is, where the specific volume is largest, as is required by mass continuity principles. It therefore follows that the velocity field is not free from divergence, the latter being in fact given by

$$\frac{1}{r} \frac{\partial c}{\partial \theta} = -\kappa \frac{\epsilon}{r^3} \sin\theta. \tag{21}$$

This expression gives (horizontal) divergence in the lower half of the figure and convergence in the upper half.

It is apparent that the pressure distribution is not radially symmetrical, but rather the pressures are lower along a given streamline on the right-hand side than on the left-hand side in the figure. This fact causes individual particles to migrate toward lower pressure in the lower half and toward higher pressure in the upper, as is indeed required by the relation that

$$\frac{dc}{dt} = -\frac{1}{\rho r} \frac{\partial p}{\partial \theta}, \tag{22}$$

derivable from (1).

According to a discussion published by the writer previously (Starr [4]), it is possible to examine the production of kinetic energy in a mechanically closed system such as the present one from the point of view of the pressure distribution and the kinematics of the existing motions. It is thus possible, in the present case, to write that

$$\frac{\partial}{\partial t} \iint \rho \frac{c^2}{2} r \, dr \, d\theta = \iint p \frac{\partial c}{\partial \theta} dr \, d\theta - \iint \frac{\partial pc}{\partial \theta} dr \, d\theta - \iint \frac{\partial}{\partial \theta} \left(c \frac{\rho c^2}{2} \right) dr \, d\theta. \tag{23}$$

Since the motion is steady the left-hand member of (23) vanishes whatever be the limits of integration. The last two terms may be converted into line integrals, so that

$$\iint p \frac{\partial c}{\partial \theta} dr \, d\theta = -\oint pc \, dr - \oint c \frac{\rho c^2}{2} dr, \tag{24}$$

the integrations on the right being performed in the counterclockwise sense. If the region is so chosen as to include the entire annular space in fig. 1, the right-hand member of (24) vanishes, so that the total kinetic-energy production then represented by the left-hand member must be zero—a fact easily verified

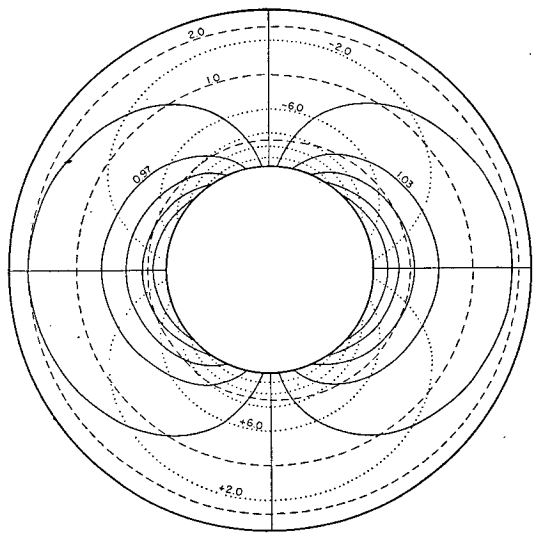

FIG. 1. Diagram of a vortex involving external heat exchanges. The air particles move in concentric circles in the counterclockwise sense with highest speeds on the right-hand side. The streamlines are not shown. The solid lines show the distribution of specific volume in c.g.s. units when the numbers are multiplied by 10^3. The dashed lines are the isobars showing the pressure distribution in millibars when 1,000.781 is added to the numbers. The dotted lines show the distribution of heat addition to system in ergs cm^{-3} sec^{-1}. In the regions near the inner boundary this heating rate far exceeds the values which could probably be realized in the atmosphere. In theory the pattern for the heating should not be symmetrical about the vertical axis of the figure, but for the values of the parameters chosen the asymmetry is insensible. The temperature distribution is not shown because it would give a pattern practically identical with that of the specific volume for the values of the parameters chosen. All lines are drawn for equal intervals of each quantity.

Unauthenticated | Downloaded 01/27/26 08:09 PM UTC

by noting the symmetrical character of the expression (16) for the pressure with respect to θ and the antisymmetric property of the expression (21) for the divergence.

If, however, a more limited region is taken, let us say the lower half, the left-hand member of (24) is positive and represents a source of kinetic energy, while the terms on the right represent kinetic-energy flow across the horizontal radii at $\theta = 0$ and $\theta = \pi$. The first and second of these latter terms are, respectively, the work done by the pressure and the advection of kinetic energy across these radii. With the aid of previous relations we find that, in this case,

$$\int\int p\frac{\partial c}{\partial \theta}\,dr\,d\theta = \epsilon\kappa^3\alpha_0(r_2 - r_1) + 2\epsilon\kappa p_0\left(\frac{1}{r_1} - \frac{1}{r_2}\right) \quad (25)$$

$$-\oint pc\,dr = 2\epsilon\kappa p_0\left(\frac{1}{r_1} - \frac{1}{r_2}\right) - \epsilon\kappa^3\alpha_0(r_2 - r_1) \quad (26)$$

$$-\oint c\frac{\rho c^2}{2}\,dr = 2\epsilon\kappa^3\alpha_0(r_2 - r_1). \quad (27)$$

It is apparent that equation (24) is fulfilled and that the total rate of kinetic-energy generation in the lower half, given by (25) is equal to the net flow of such energy into the upper half as given by (26) and (27). The lower half thus represents a region of direct transformation of heat energy into kinetic energy which is then transferred to the upper half and reconverted directly into heat energy again. Part of the heat transfer is thus accomplished through the flow of kinetic energy.

It is also possible to compute the net flow of internal heat energy by advection out of the lower half. This quantity is

$$-\oint c_v c\rho T\,dr = 2c_v A\epsilon\kappa p_0\left(\frac{1}{r_1} - \frac{1}{r_2}\right) - c_v A\epsilon\kappa^3\alpha_0(r_2 - r_1). \quad (28)$$

The total outflow of internal and kinetic energy from the lower half may be obtained by adding the expressions (26), (27), (28) and is found to be

$$\epsilon\kappa^3\alpha_0(1 - c_v A)(r_2 - r_1) + 2\epsilon\kappa p_0(1 + c_v A)(r_1^{-1} - r_2^{-1}). \quad (29)$$

According to the principle of conservation of energy this should be equal to the heat gain in the lower half. Integration of (20) leads to precisely the expression (29).

5. Meteorological applications

The specific model which has been calculated cannot be considered as a direct analogue of any actual atmospheric system, although it probably exhibits certain features in common with such systems. Its purpose in this respect is merely suggestive. Thus one noteworthy feature of the model is that the conversion of heat energy into kinetic energy in the lower half takes place directly without the mediation of gravity. In this respect the conversion process is similar to that occurring in a heat engine. One is therefore tempted to raise the question whether this may not be largely true also of the general circulation of the atmosphere. Modern observations (see Vuorela [5], for example) do not seem to fit very well with the notion that the energy for the general circulation is provided by large-scale vertical convective cells of the Hadley type in which gravity would of course play an important role. Other mechanisms for generating kinetic energy are therefore of theoretical interest.

The fundamental differential equation (4) needs various modifications in order to make it more applicable to large-scale atmospheric motions. For such purposes it is important to observe the following:

a. An equation of the form of (4) may be used for the solution of problems involving nonsteady flow, provided that the flow patterns move uniformly without change of shape, so that the motions relative to appropriate moving coordinates are steady.

b. The introduction of friction[3] proportional to either the velocity or the momentum into equations (1) does not destroy the linearity of (4).

c. The introduction of Coriolis forces depending upon a constant Coriolis parameter into (1) leaves equation (4) unaltered. For large atmospheric systems it is however essential to use a variable Coriolis parameter in which case the linearity of (4) is still preserved.

d. If we imagine that the axis of y is taken as vertical, the introduction of gravity renders equation (4) nonlinear, although if the total range of variation of α is small, the equation can probably be successfully linearized.

It would be of considerable interest to apply the method discussed in this paper to large-scale horizontal flow patterns resembling those found in the general circulation of the atmosphere, provided that the necessary adaptations can be made. These are mainly the adaptation to flow in a spherical stratum, the introduction of ground friction and the introduction of Coriolis forces. The most general attempt of this sort might encompass an entire hemisphere, in which case it would be necessary to introduce stream-

[3] The incorporation of friction proportional to the momentum in the model previously computed leads to a general solution

$$\alpha = \chi_1(r) + r^{-3}\chi_2(\theta) - 2k\theta/3\kappa.$$

Here χ_1 and χ_2 are arbitrary functions, while k is the coefficient of friction. Due to the presence of the last term on the right it is evident that there must be at least one discontinuity in density or branch line emanating from the origin. From angular-momentum considerations it is clear that the flow cannot be reëntrant, although it could be thought of as taking place in a Riemann surface of more than one sheet.

Unauthenticated | Downloaded 01/27/26 08:09 PM UTC

line patterns having certain over-all characteristics, namely: (*a*) balance of frictional torques upon the earth's surface, and (*b*) suitable "tilt" of trough- and ridge-lines and other features to ensure a proper meridional flow of angular momentum as discussed previously by the writer (Starr [3]). In all probability the computations involved in such a scheme would be too tedious to perform without the aid of modern high-speed computing machines. Since such equipment is now becoming available for meteorological purposes, a project of this character may be justified.

The merit of this approach would depend upon whether the distribution of the necessary heat sources and heat sinks is one compatible with the general available information concerning the radiative heat balance of the earth.

Acknowledgment.—The writer wishes to express his gratitude to Dr. E. N. Lorenz and Mr. R. C. Cowen for many valuable discussions pertaining to the subject of this paper.

REFERENCES

1. Forsyth, A. R., 1929: *A treatise on differential equations*. 6 ed., London, MacMillan and Company, 424 pp.
2. Hicks, B. L., 1949: On the characterization of fields of diabatic flow. *Quart. appl. Math.*, **6**, 405–416.
3. Starr, V. P., 1948: An essay on the general circulation of the earth's atmosphere. *J. Meteor.*, **5**, 39–43.
4. ——, 1948: On the production of kinetic energy in the atmosphere. *J. Meteor.*, **5**, 193–196.
5. Vuorela, L. A., 1948: Contribution to the aerology of the tropical Atlantic. *J. Meteor.*, **5**, 115–117.
6. Webster, A. G., 1933: *Partial differential equations of mathematical physics*. Leipzig, B. G. Teubner, 440 pp.

Unauthenticated | Downloaded 01/27/26 08:09 PM UTC